👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

# Area of Regular Polygons

Lesson 5-9 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can find the area of a regular polygon by decomposing it into triangles.**

**Language Objective: I can explain my steps using the words regular polygon, decompose, triangle, and composite.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Regular Polygon | Decompose | Triangle | Composite | Formula

1. A polygon with all sides and all angles equal is a _____.

2. To break a shape into smaller shapes is to _____ it.

3. A three-sided polygon is a _____.

4. Made of two or more shapes combined, a figure is _____.

5 A math rule written with symbols is a _____ .

## Write About the Math

**How much will the gazebo flooring cost?**
*¿Cuánto costará el piso del quiosco?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Regular Polygon** — *Polígono regular*
- ☐ **Decompose** — *Descomponer*
- ☐ **Triangle** — *Triángulo*
- ☐ **Composite** — *Compuesto*
- ☐ **Formula** — *Fórmula*

**Model:** *We decompose the hexagon into triangles because we already know how to find a triangle's area.* — Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

### Start

Complete the frames. Use the word bank.
*Completa las oraciones. Usa el banco de palabras.*

- **One triangle's area is ___ sq ft. The octagon has ___ triangles, so total area = ___ sq ft. The cost is ___ x $3 = $___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Regular Polygon
2. **Notes 2:** Decompose
3. **Notes 3:** Triangle
4. **Notes 4:** Composite
5. **Notes 5:** Formula
6. **Watch for this mistake:** *A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area — or adding the number of triangles instead of multiplying. Remember: total area = (one triangle's area) × (number of sides), not (one triangle's area) + (number of sides).*
7. **Try It 1:** C. 28 $in^2$ — *One triangle = ½ × 4 × 2.8 = 5.6 $in^2$. A pentagon has 5 triangles, so total = 5 × 5.6 = 28 $in^2$.*
8. **Try It 2:** D. 172.8 sq ft — *An octagon has 8 sides, so it splits into 8 triangles. One triangle = ½ × 6 × 7.2 = 21.6 sq ft. Total = 8 × 21.6 = 172.8 sq ft.*